

Sammy Abdella

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EDUCATION

Columbia University in The City of New York

B.S. Computer Science

SKILLS

Technical: Python, Git, Linux, REST, PostgreSQL, C++, Java, JavaScript, SQL, HTML/CSS, Ruby, OAuth2, JWT, Docker, Kubernetes

Languages: French, Arabic, English

EXPERIENCE

Web Developer Intern | Columbia College IT Dept | New York, NY

08/2025 – Present

- Migrated 43 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, building/refreshing site themes with HTML/CSS, running regression tests, improving average page load times by ~35%, and removing 100% of deprecated modules in production
- Integrated a Django, PostgreSQL, and RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Backend Software Developer Intern | Casca | Remote

06/2024 – 09/2024

- Built async webhook pipeline with Celery/Redis, reducing settlement processing lag ~25% and cutting duplicate payout incidents by 70% through idempotency and retry controls
- Implemented OpenTelemetry tracing and structured logging across services; shortened incident triage time ~30% and improved root-cause identification during outages for on-call engineers
- Strengthened transaction services with RBAC, audit trails, and secrets rotation; reduced unauthorized-access alerts ~15%, improved audit readiness, and made large deployments more reliable under peak traffic

Research Intern | Columbia University AI Laboratory | New York, NY

01/2025 – 09/2025

- Led the development for gAyl, an AI tool for LGBTQ+ suicide prevention, designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication (co-author): <https://doi.org/10.7916/z6vc-d867>

Amazon SWE Intern

Jun 2023 - Aug 2023

- Built a Java + DynamoDB internal admin service that consolidated three legacy CLIs into a single REST API used by an on-call team of ~25 engineers. Cut average incident response time on a recurring class of tickets by ~40%.
- Wrote and shipped a CloudWatch alarm framework plus runbook updates for a backend pipeline, reducing repeat noisy alerts by ~25% over the rotation.
- Improved a batch reconciliation job by introducing pagination and parallel processing in Java. Reduced job runtime from ~6 hours to ~1.5 hours on production scale data.

Projects

AI-Powered Exam Tutor | Python, Django, PostgreSQL, Redis, Celery, Docker

09/2025 – 01/2026

- Built a web app that generates timed practice exams, stores attempts, and produces graded reports with explanations and targeted remediation suggestions
- Deployed with Columbia University's Learning Center to support 50+ students across Data Structures, Advanced Programming, and Multivariable Calculus exam prep